

BANGLADESH UNIVERSITY OF PROFESSIONALS

DEPARTMENT OF INFORMATION AND COMMUNICATION TECHNOLOGY

Digital Communication

Introduction to Digital Communication, Introduction to Communication Channels, Mathematical Models and Characteristics of Communications Channels (week 1)

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Contents

- Introduction to Digital Communication
- Introduction to Communication channel
- Characteristics of a Communication Channel
- Mathematical model of a Communication Channel

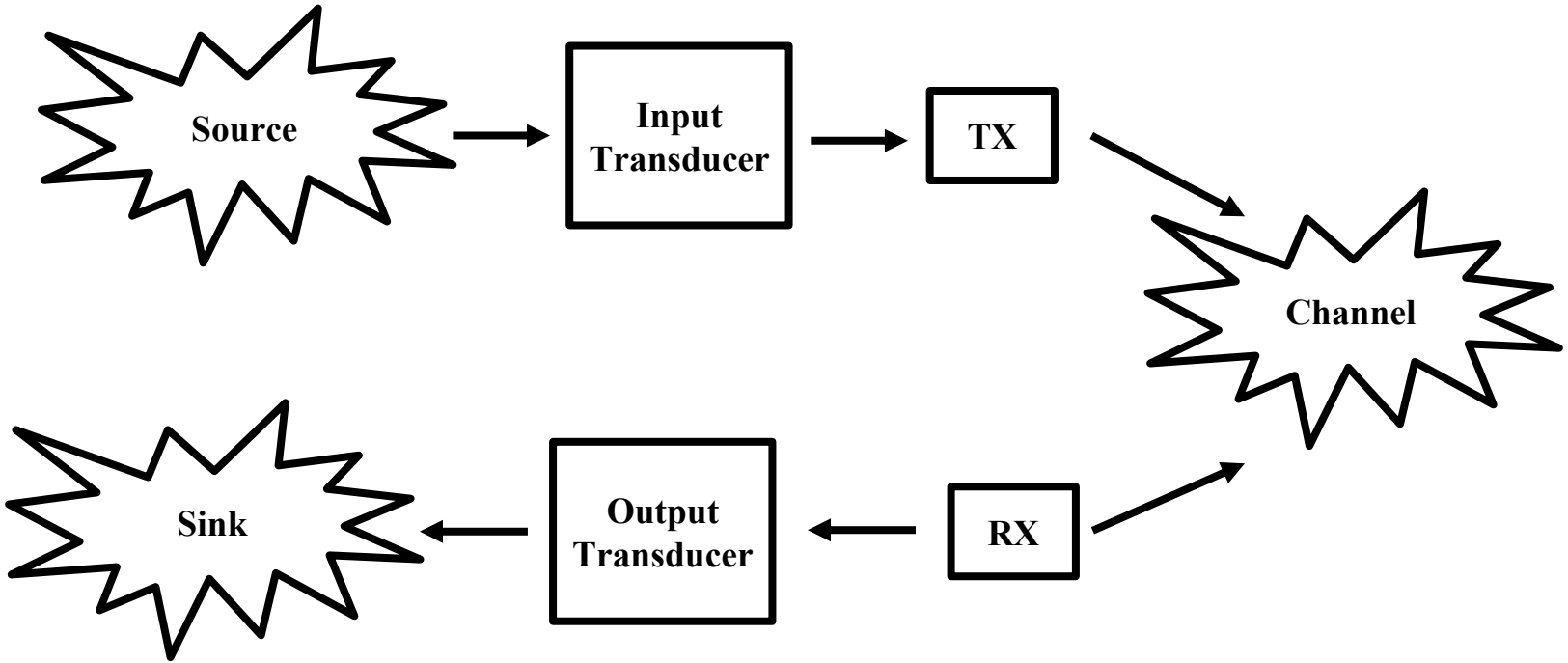
"Communication has been one of the deepest needs of the human race throughout recorded history."

- ***Digital Communication***

  Digital communication systems, by definition, are communication systems that use such a digital sequence as an interface between the source and the channel input (and similarly between the channel output and final destination).

  A digital sequence is a sequence made up of elements from a finite alphabet (e.g., the binary digits {0, 1}, the decimal digits {0, 1,... , 9} , or the letters of the English alphabet) . The binary digits are almost universally used for digital communication and storage, so we only distinguish digital from binary in those few places where the difference is significant.

- ***Block Diagram***



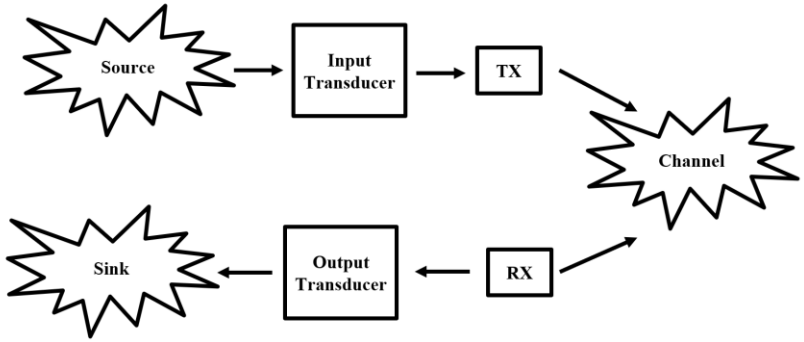
- ***Purpose***

Send Information from Information Source to Information Sink

Take Information from TX to the end of RX.

Waveforms from TX is replicated to the end of RX.

- ***Block Diagram***

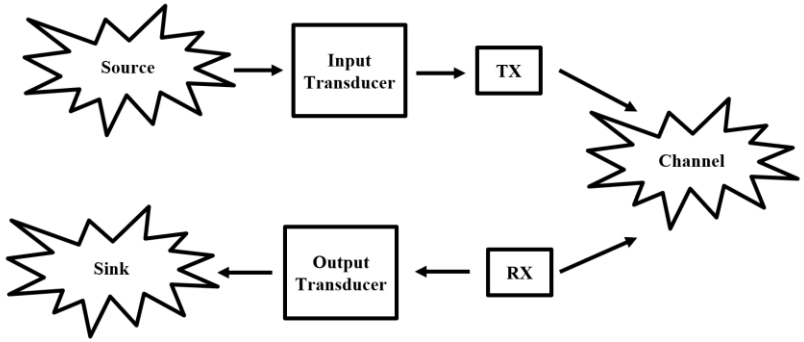


❑ ***Source***

Any real-world Activity/system that can produce information.

- Typing
- Speaking
- Reciting

❑ ***Block Diagram***



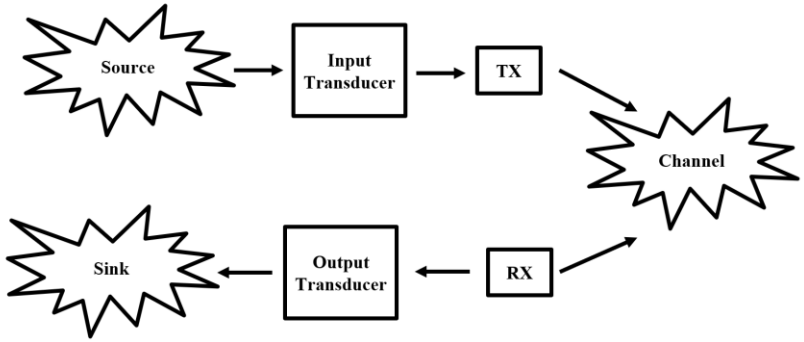
❑ *Input Transducer*

Any device where one end interface will be the source and other end produces signal that we can use in the communication system.

Most of the time generates baseband signals.

- Microphone.
- Keyboard.
- Camera.

❑ *Block Diagram*



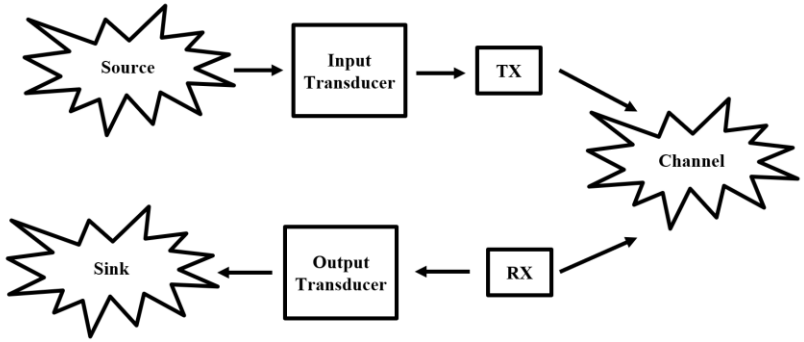
"Transfers the real world to the world of signals."

❑ ***Output Transducer***

It is the inverse of the input transducer where the signals (of voltages/currents) are translated in the form that is understandable by the sink.

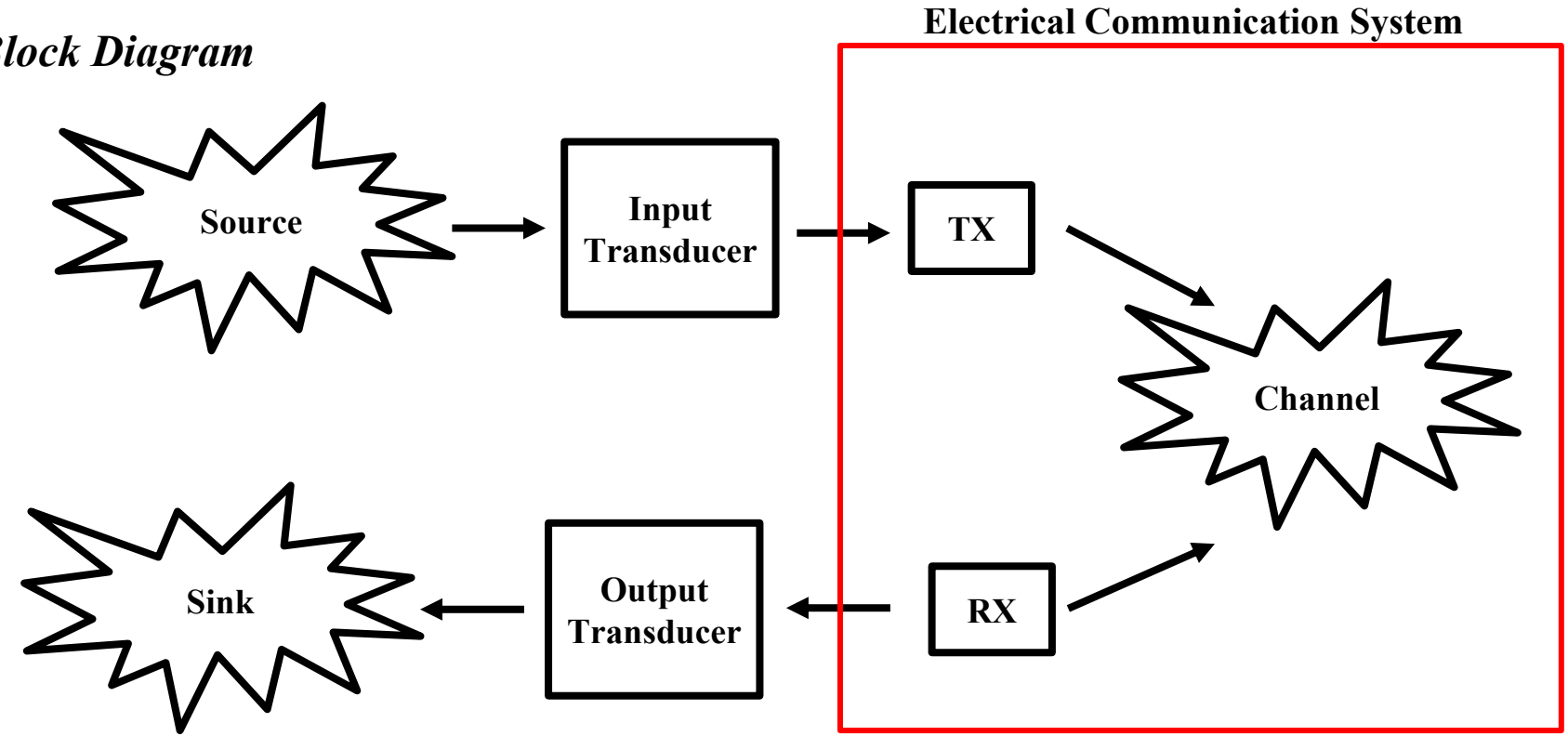
- Loudspeakers.
- Buzzers.
- Monitors.

❑ ***Block Diagram***



"It transfers the world of electrical signals back into the real world."

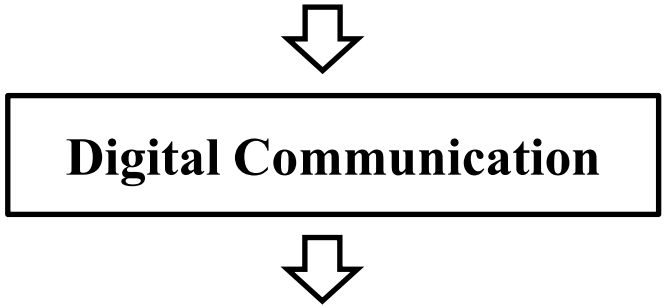
- ***Block Diagram***



- ***Example***

Speaking => Microphone producing voltages and currents

Source — Input Transducer



=> Voltages and Currents are converted to the acoustic sound

Output Transducer

=>Loudspeaker

Sink

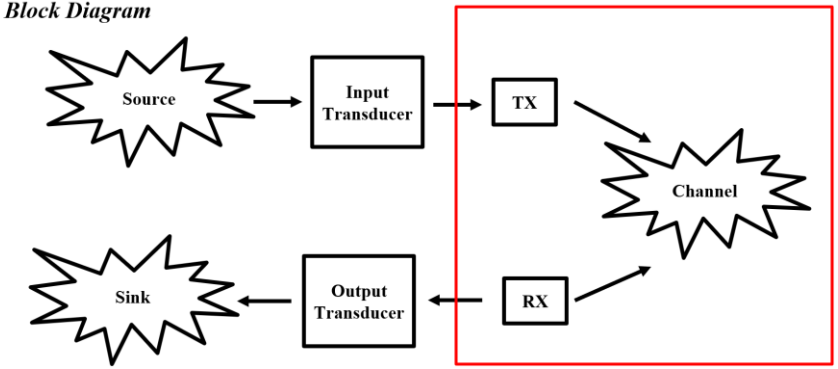
- ***Digital Communication***

TX – Transmitter

Channel

RX – Receiver

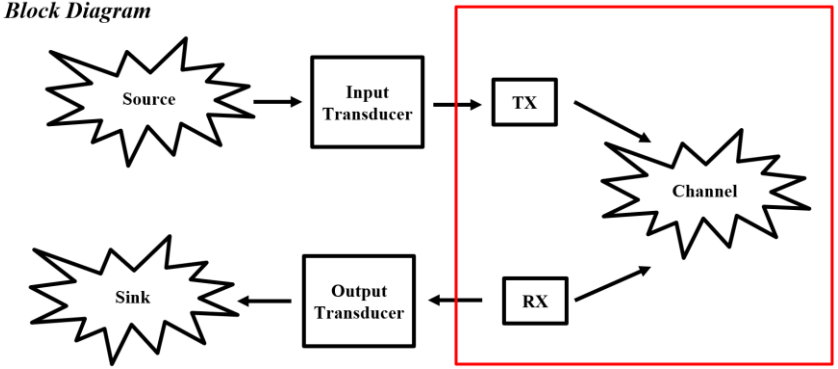
- *Block Diagram*



❑ *Transmitter*

Transmitter converts electrical signals into another set of signals/waveforms that are suitable for the transmission over a particular channel.

❑ *Block Diagram*



There are some operations to be performed.

- A low-pass or baseband signal occupies a frequency range from near 0 Hz up to a maximum cutoff frequency (f_c).
- Raw information like audio, video, or data.
- Telephone voice frequency is up to 4 kHz, Human voice up to 255Hz.

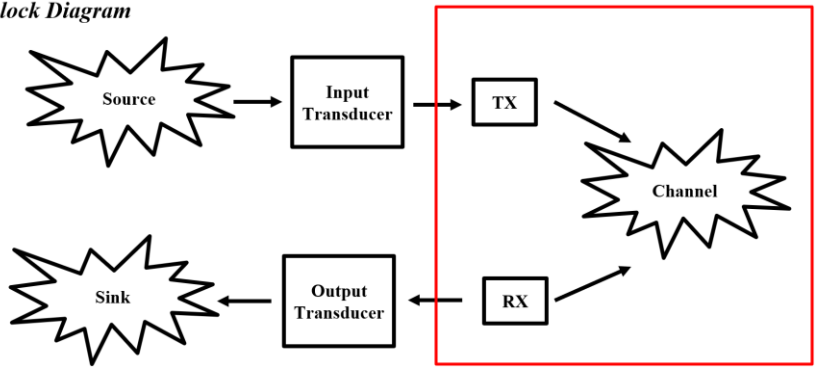
- ***Transmitter Operation:***

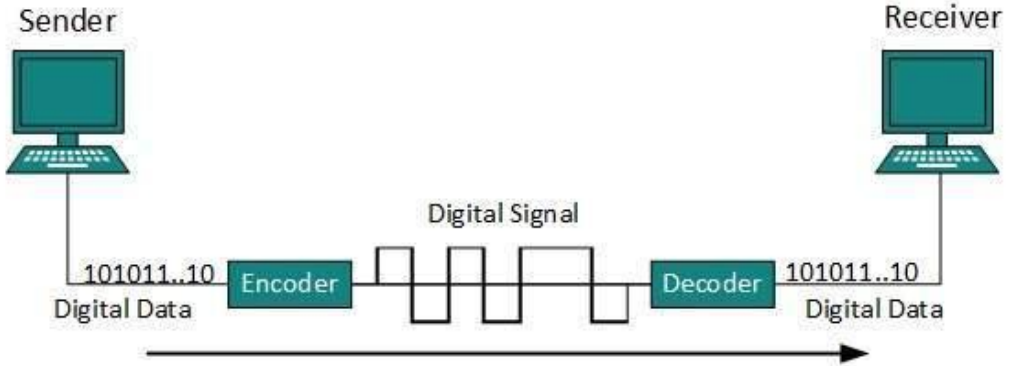
Signal Formatting

Transforming, encoding, and structuring information into a specific format that the transmission medium and the receiver can understand.

- Companding.
- Compression.
- Thresholding.

- ***Block Diagram***

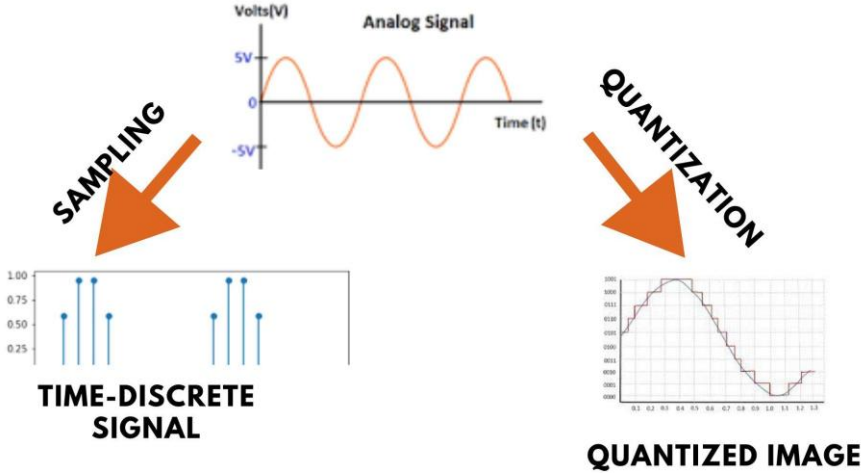




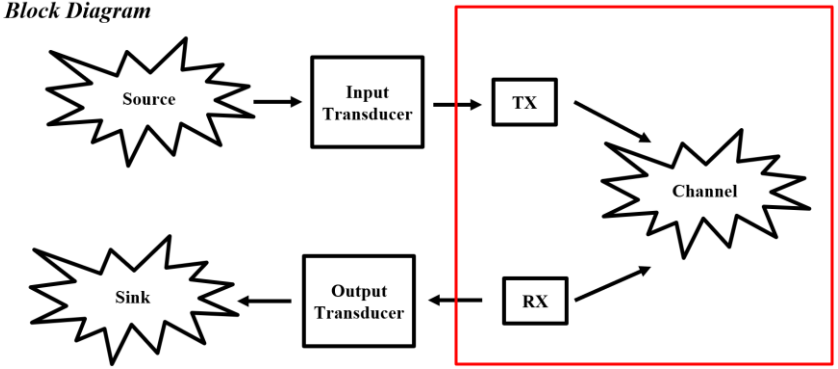
❑ ***Transmitter Operation:***

Sampling and Quantization

Foundational transmitter operations in digital communication systems, forming the core of Analog-to-Digital Conversion (ADC)



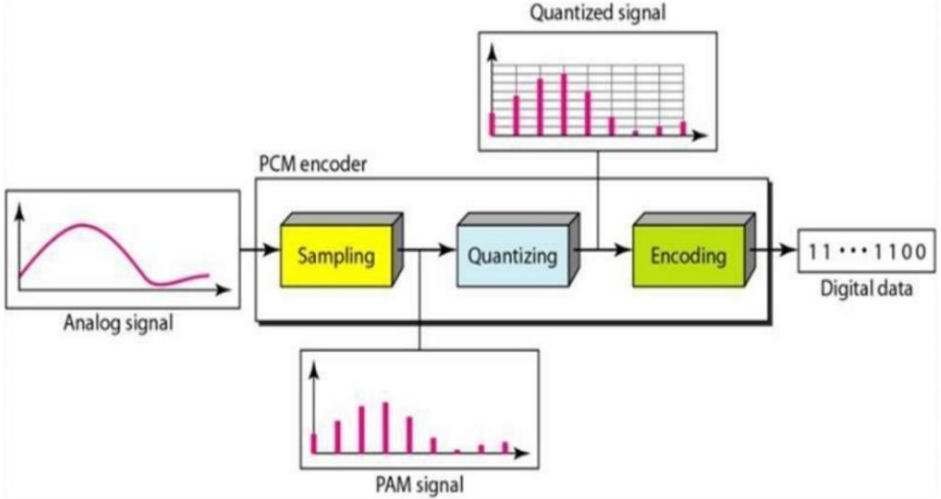
❑ *Block Diagram*



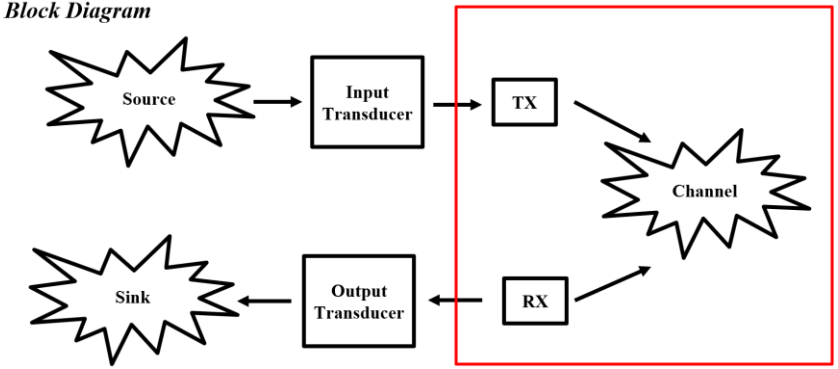
- ***Transmitter Operation:***

Source Encoding (Quantization Mapping)

Maps a large, continuous set of amplitudes to a smaller, finite set of discrete levels.



- *Block Diagram*

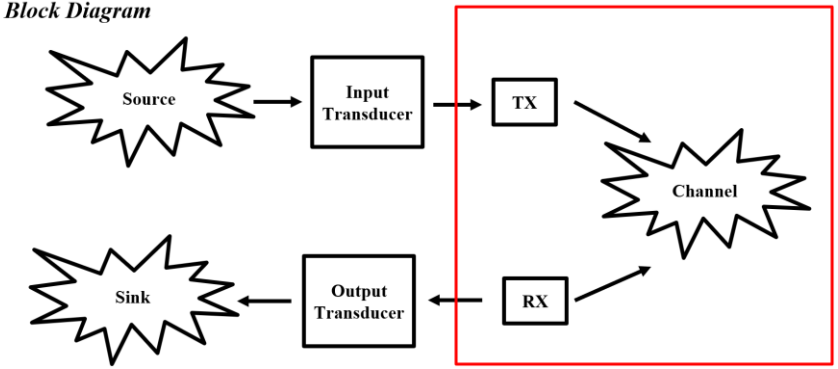


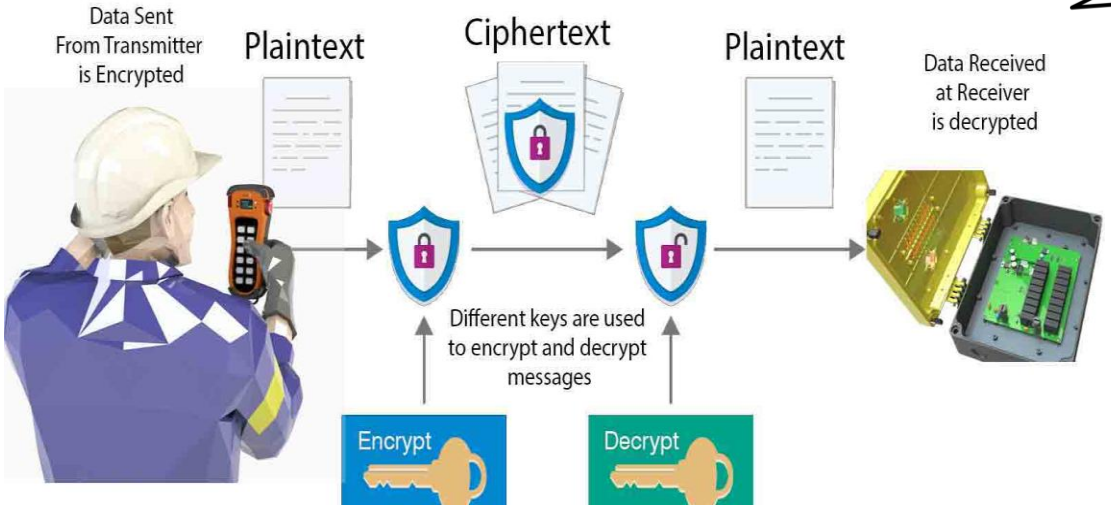
- ***Transmitter Operation:***

Encryption

The process of converting plaintext data into ciphertext using cryptographic protocols

- ***Block Diagram***



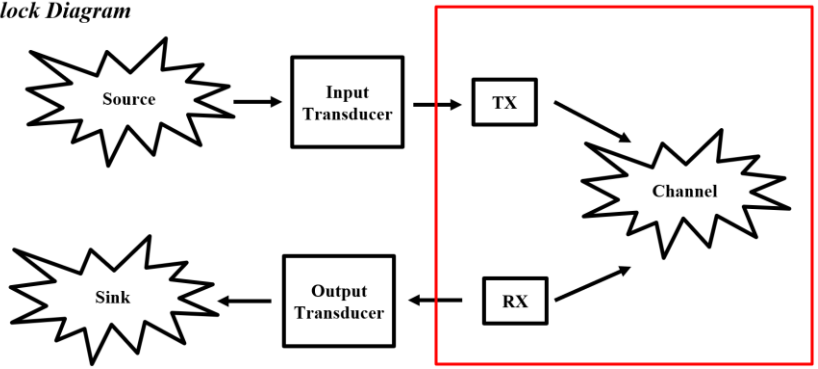


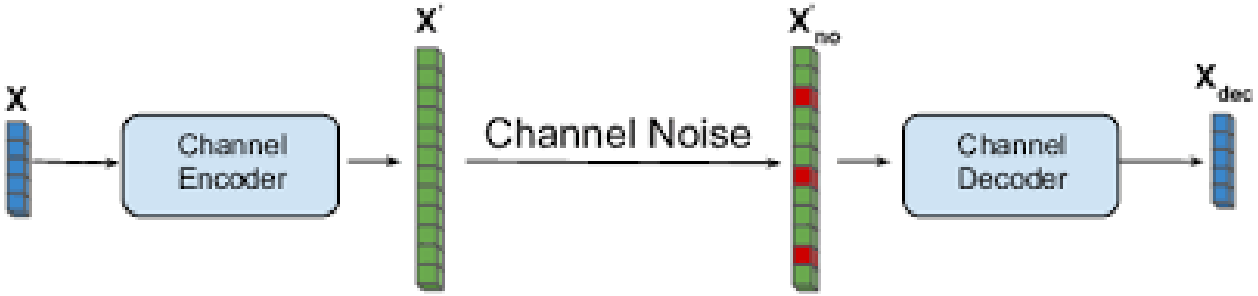
- ***Transmitter Operation:***

Channel Encoding

process that adds redundant bits to data, enabling the receiver to detect and correct errors caused by noise, interference, or fading during transmission.

- *Block Diagram*

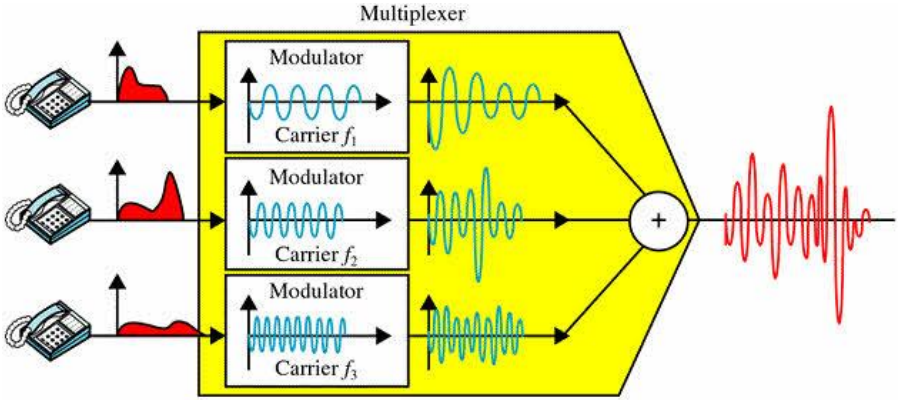




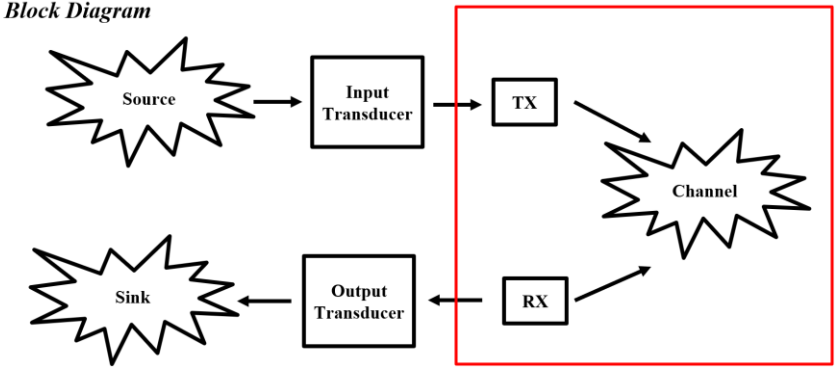
- ***Transmitter Operation:***

Multiplexing

A technique that combines multiple analog or digital signals into one, allowing them to be transmitted simultaneously over a single shared medium - to maximize bandwidth efficiency



- *Block Diagram*

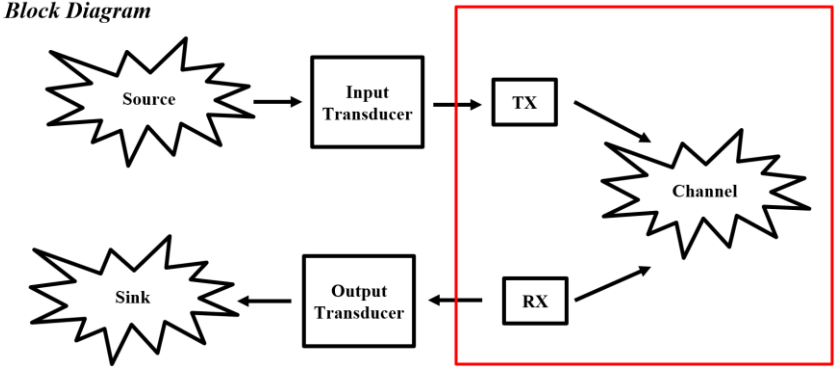


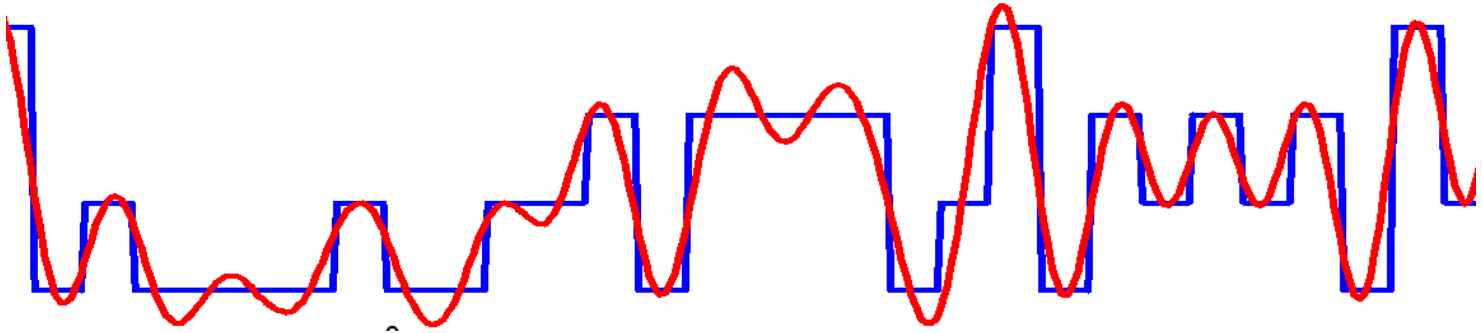
- ***Transmitter Operation:***

Pulse Shaping

A critical transmitter modifies the shape of pulses (e.g., shaping rectangular pulses into smoother waveforms like raised-cosine) before modulation

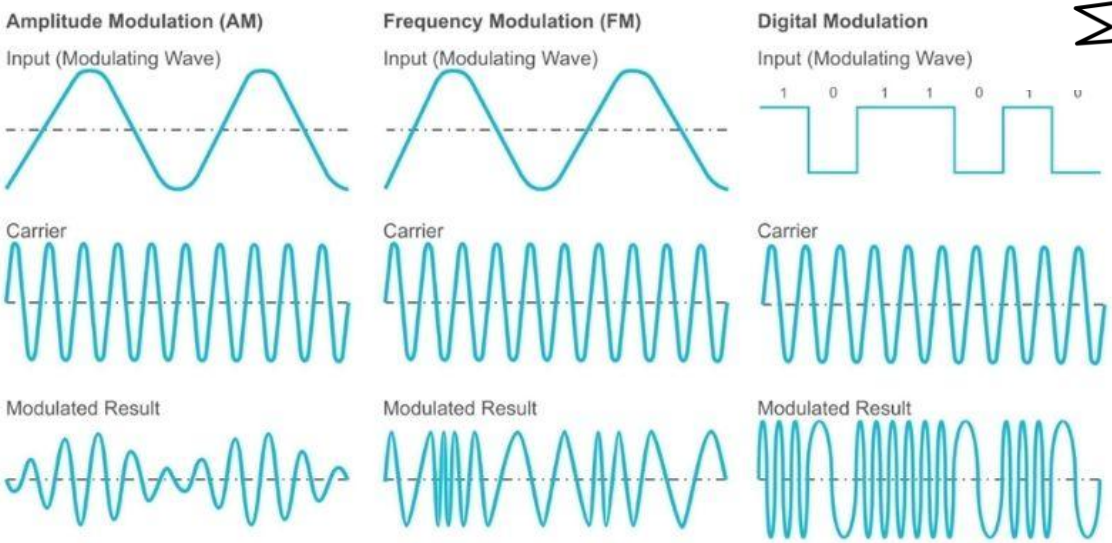
- *Block Diagram*



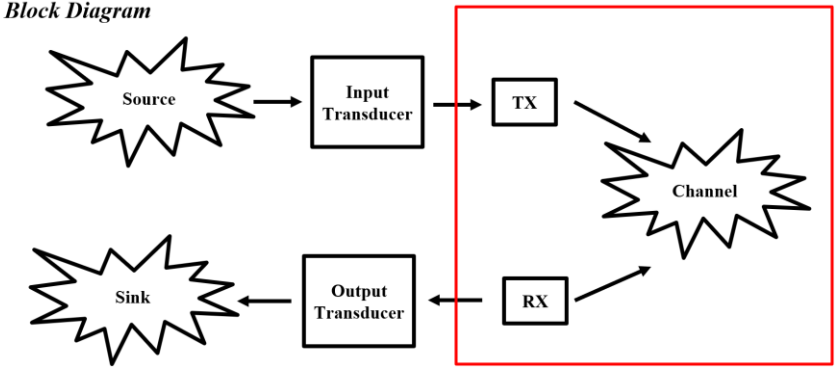


- ***Transmitter Operation: Modulation***

It is fundamental operation of superimposing a low-frequency information signal (voice, data, video) onto a high-frequency carrier wave



- ***Block Diagram***

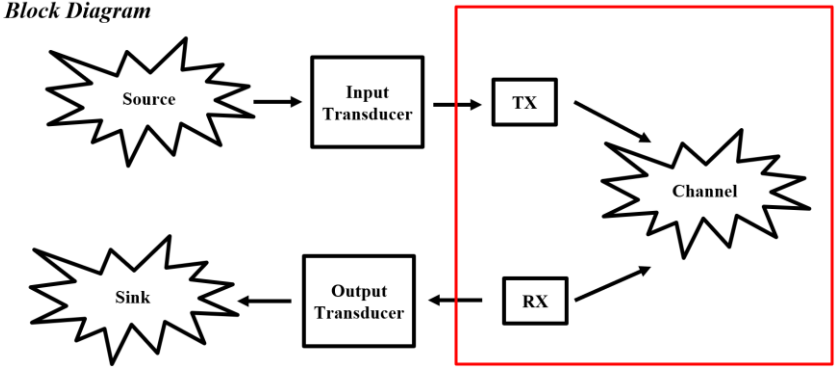


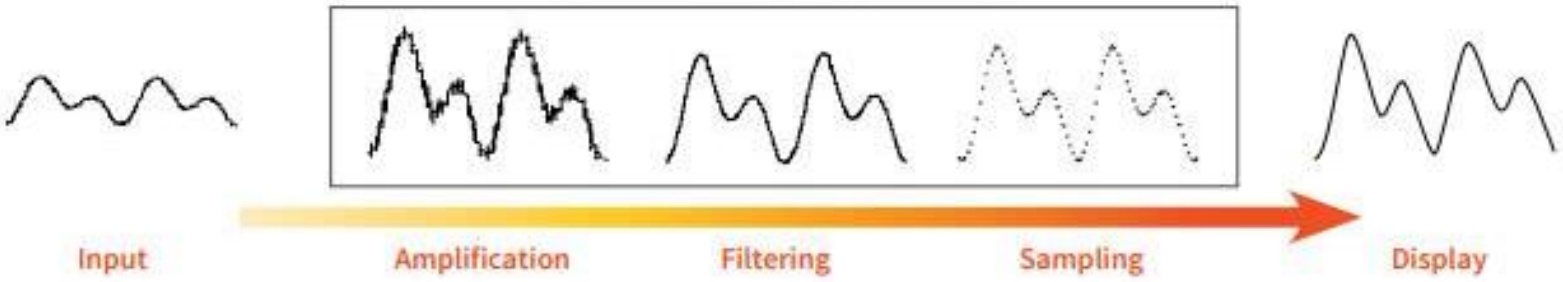
- ***Transmitter Operation:***

Amplification & Filtering

Processes that increase signal strength and remove unwanted noise, ensuring accurate data acquisition from sensors.

- ***Block Diagram***



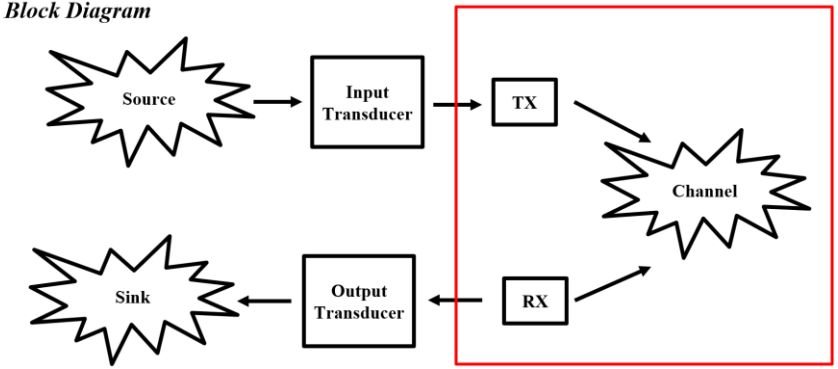


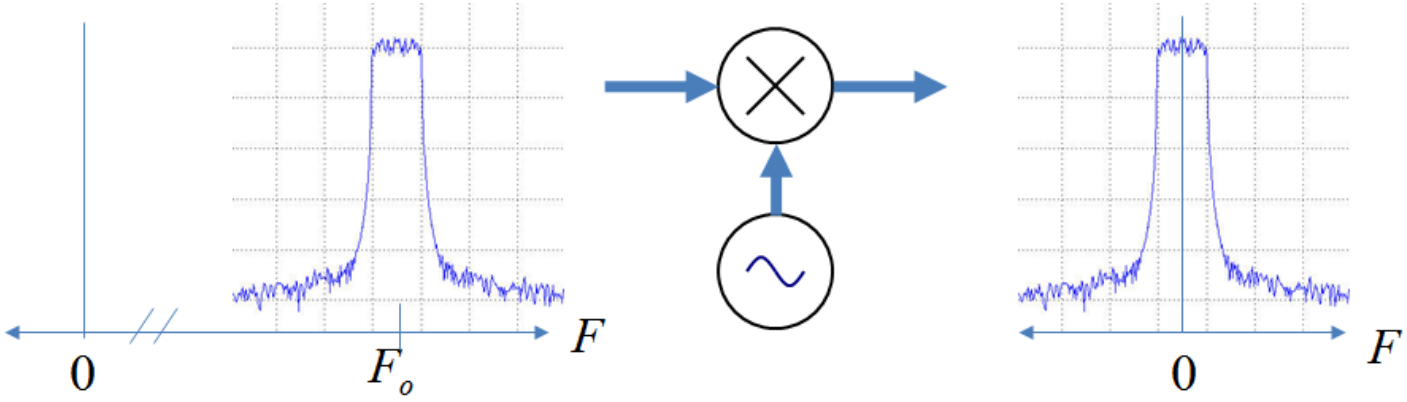
- ***Transmitter Operation:***

Frequency Translation

mechanism for moving a low-frequency, baseband information signal (like voice or data) up to a much higher, radio-frequency (RF) carrier frequency for transmission.

- *Block Diagram*





Introduction to Digital Communication

- ***Transmitter Operations:***

  Signal Formatting

- ✓ ***Sampling and Quantization***
- ✓ ***Source Encoding (Quantization Mapping)***

  Encryption

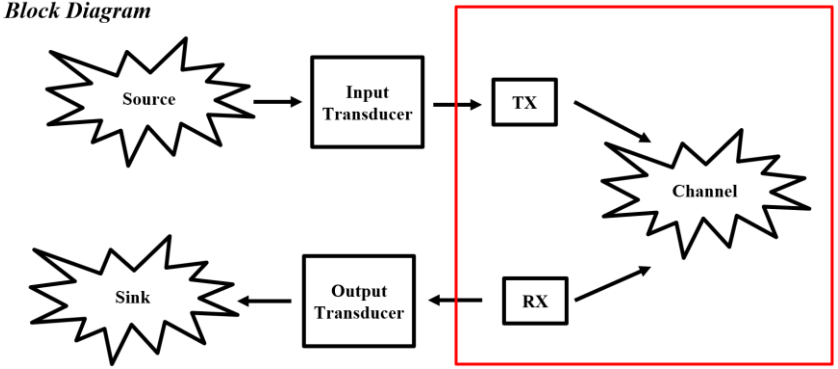
  Channel Encoding

- ✓ ***Multiplexing***
- ✓ ***Pulse Shaping***
- ✓ ***Modulation***

  Amplification & Filtering

- ✓ ***Frequency Translation***

- ***Block Diagram***



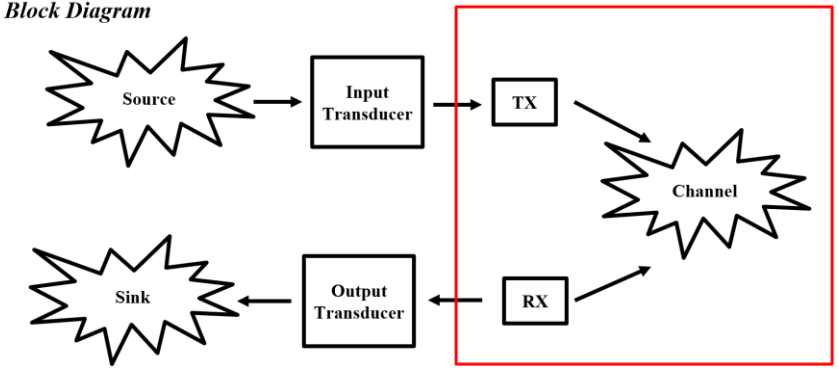
- ***Channel:***

Physical medium used to send signal from transmitter to receiver; have physical properties.

Instead of investigating practically, abstract the channel to mathematical model.

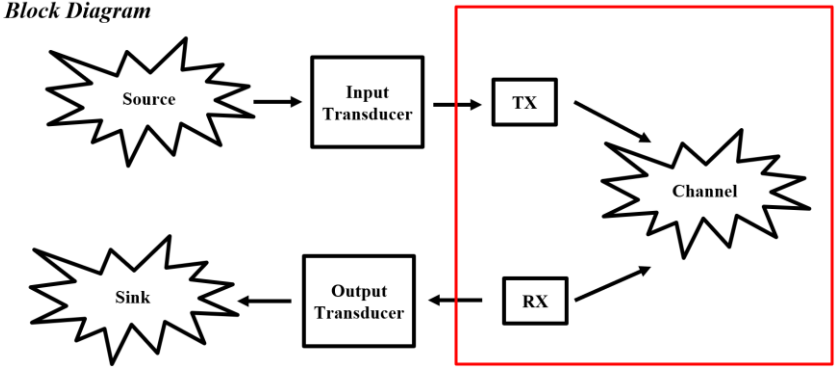
Investigate the characteristics of the channels.

- *Block Diagram*



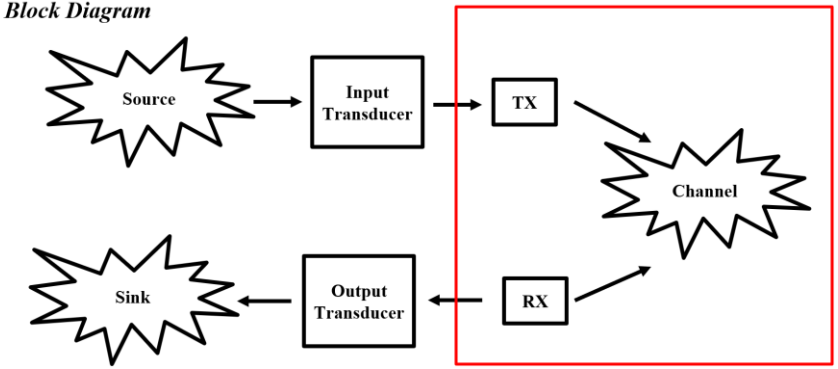
- ***Channel:***
- Wired
  - Optical (transmits light)
    - Optical fiber.
  - Electrical (transmits voltages and currents)
    - Twisted pair.
    - Co-axial cables.
    - Waveguide.

- *Block Diagram*



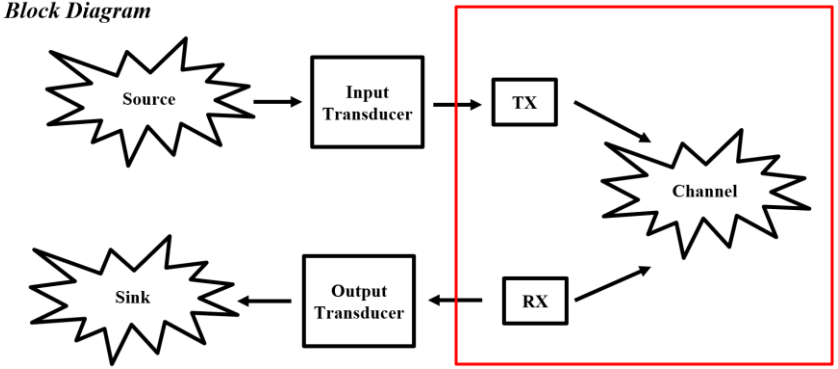
- ***Channel:***
  - Wireless
    - Optical (Through Space)
      - Infrared Channel.
      - Electromagnetic Channel.
        (waves of the optical portion of the spectrum)
    - Radio
    - Acoustic.
      (Underwater communication- electromagnetic wave do not propagate well under water)

- *Block Diagram*



- ***Channel:***
  - Wired
  - Wireless
  - Whatever we put in the channel and whatever comes out of the channel is different.

- *Block Diagram*

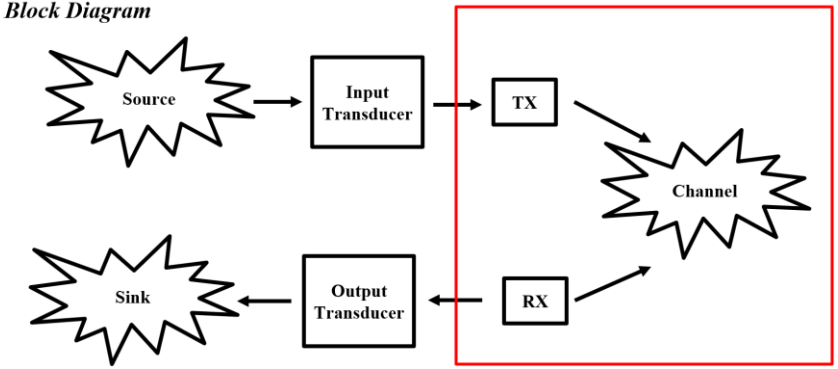


"There is Distortion everywhere"

- ***Mathematical Models of Channel:***

1. Additive Noise Channel.
2. Linear Time-invariant Filter Channel.
3. Linear time-variant Filter Channel.

- *Block Diagram*



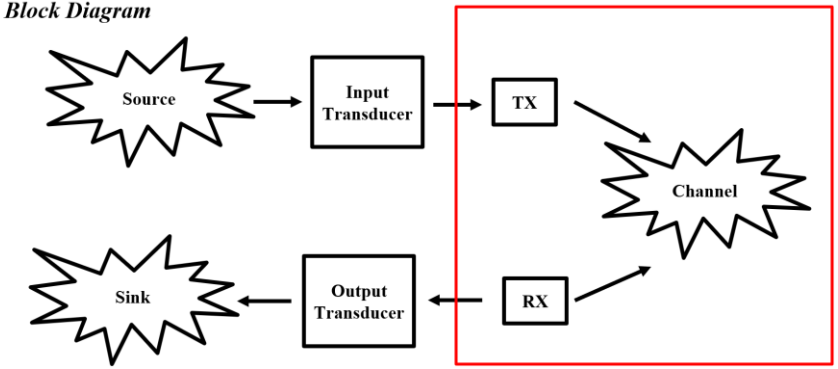
"There is Distortion everywhere"

- ***Channel Distortion***

1. Additive Distortion: output of the channel contains other signals beyond what is caused by input. (noise, interference)

2. Non-additive Distortion: spectral modification, filtering, non-linear distortion

- *Block Diagram*



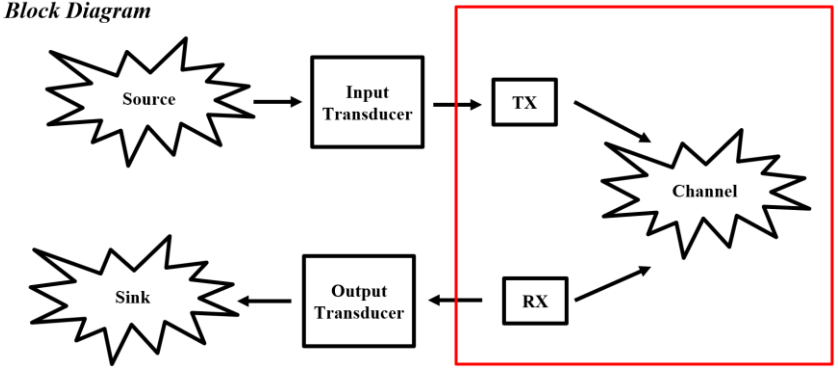
"There is Distortion everywhere"

- ***Receiver***

Recovers original signal that was passed to the transmitter

Most Crucial and Complicated Task.

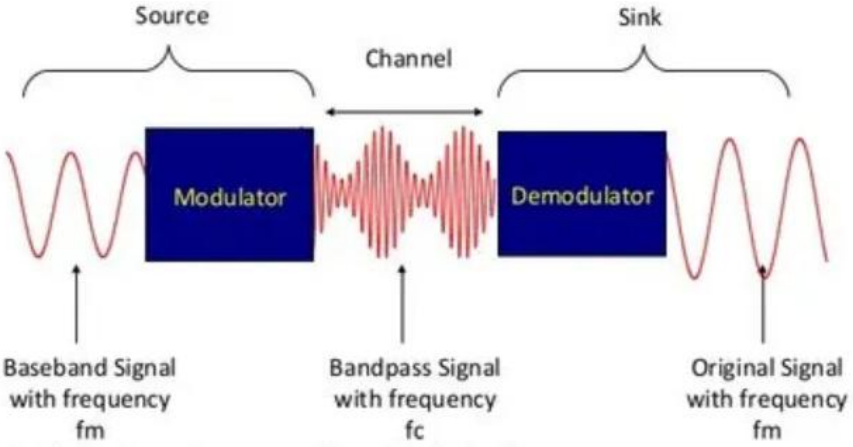
- *Block Diagram*



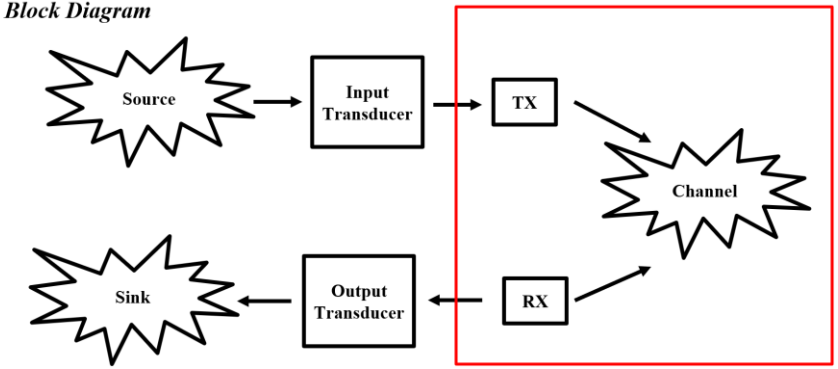
Need to do all the complementary things that the transmitter do.

Challenges include- the signal can be heavily distorted.

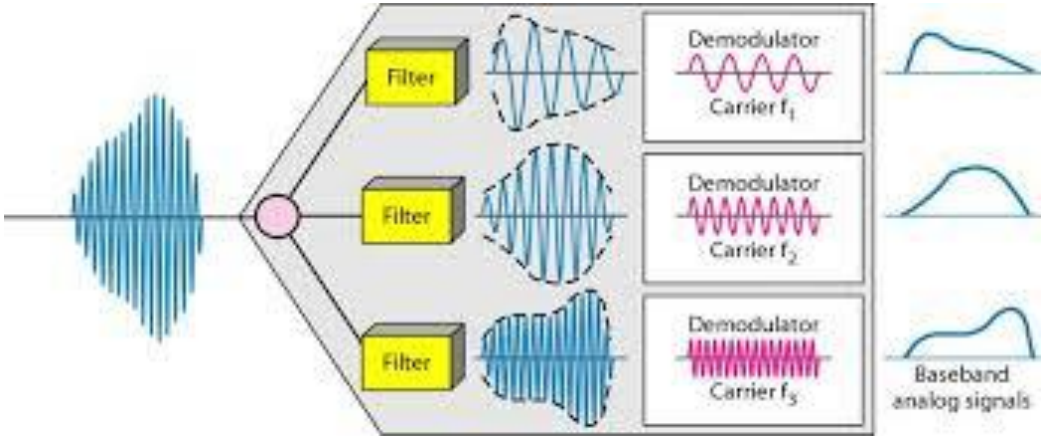
- ***Receiver Operation:***
- ✓ ***Frequency translation***
- ✓ ***Amplification and Filtering***
- ✓ ***Demodulation***



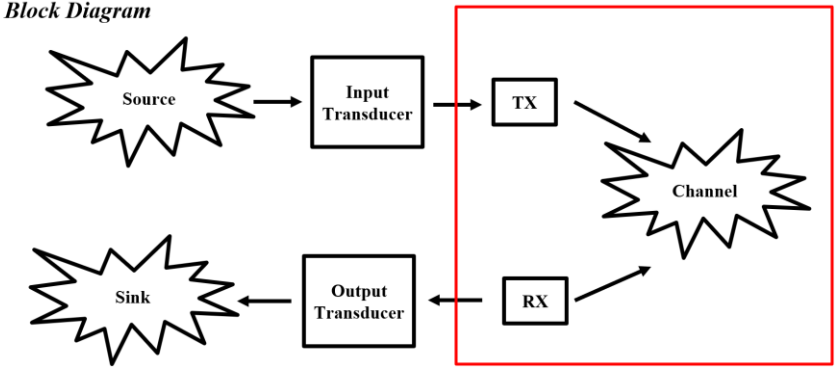
- ***Block Diagram***



- ***Receiver Operation:***
- ✓ ***Frequency translation***
- ✓ ***Amplification and Filtering***
- ✓ ***Demodulation***
- ✓ ***Demultiplexing***

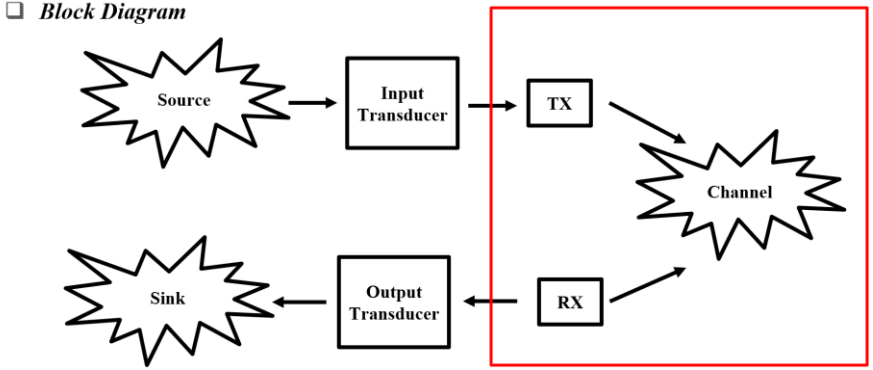


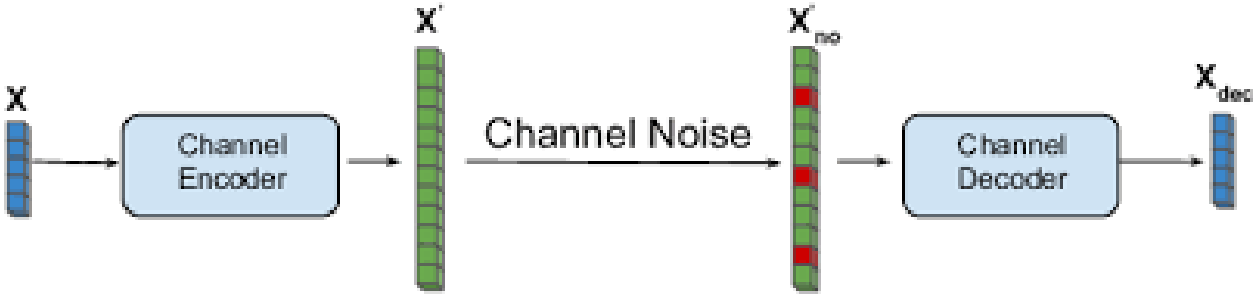
- ***Block Diagram***



- ***Receiver Operation:***
- ✓ ***Frequency translation***
- ✓ ***Amplification and Filtering***
- ✓ ***Demodulation***
- ✓ ***Demultiplexing***
- ***Decoding and Error Correction***

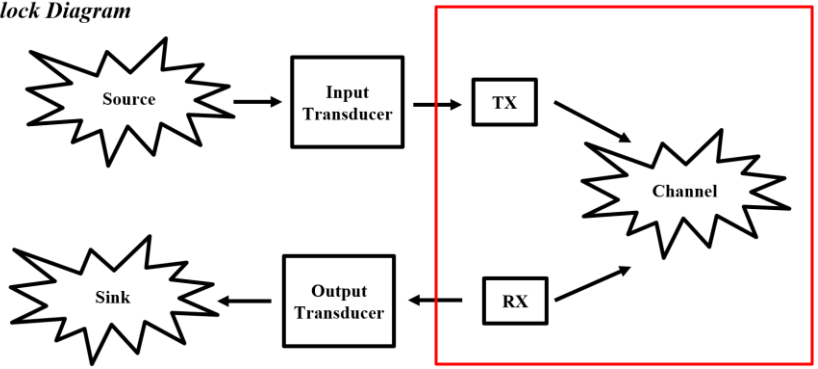
- ***Block Diagram***

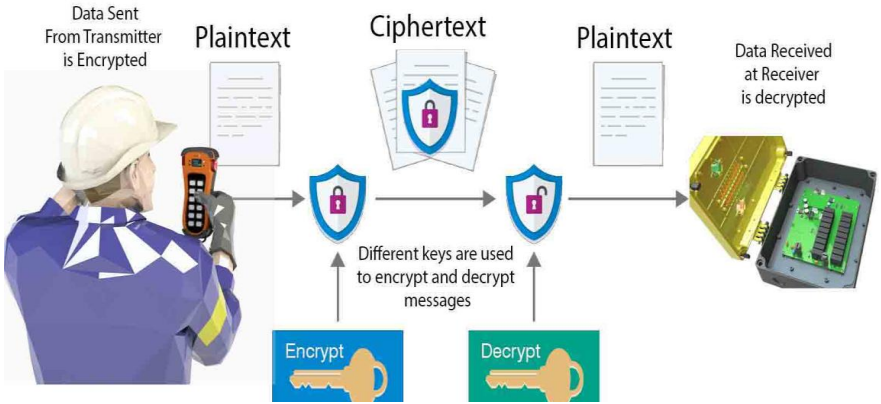




- ***Receiver Operation:***
- ✓ ***Frequency translation***
- ✓ ***Amplification and Filtering***
- ✓ ***Demodulation***
- ✓ ***Demultiplexing***
- ***Decoding and Error Correction***
- ***Deciphering***

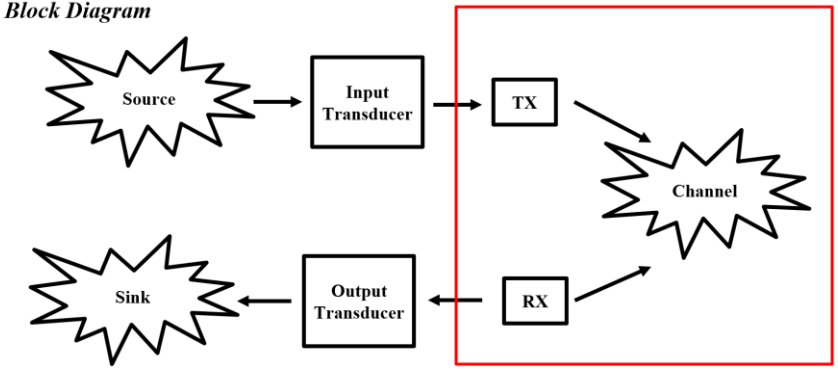
- ***Block Diagram***

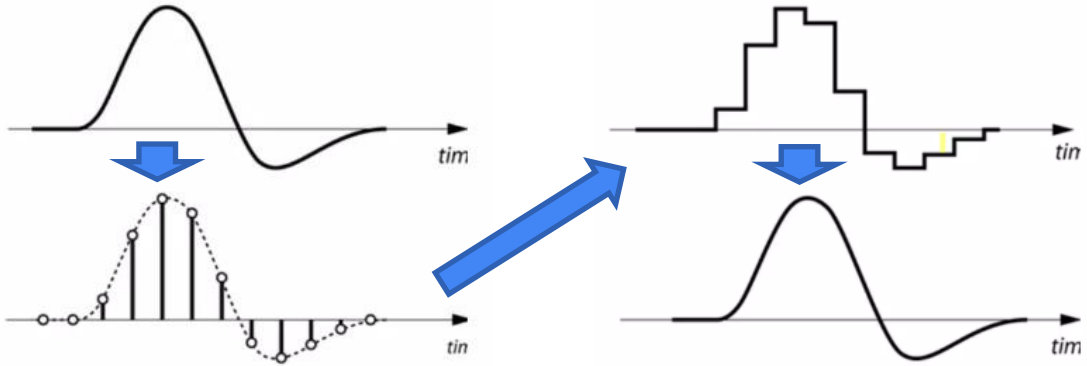




- ***Receiver Operation:***
- ✓ ***Frequency translation***
- ✓ ***Amplification and Filtering***
- ✓ ***Demodulation***
- ✓ ***Demultiplexing***
  Decoding and Error Correction
  Deciphering
- ✓ ***Reconstruction***

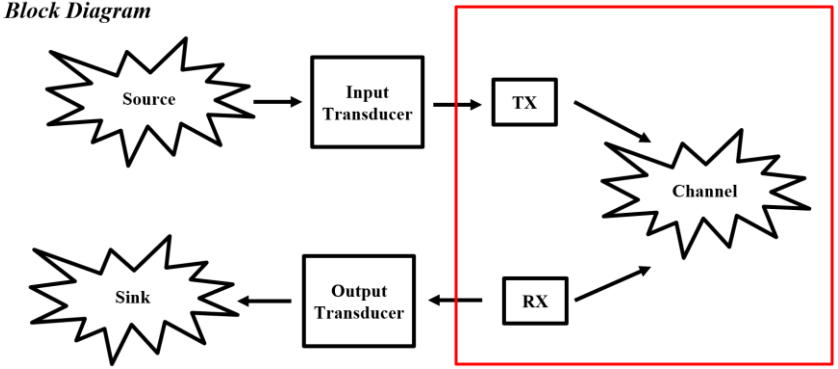
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- ***Deciphering***
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- ✓ ***Formatting***

- ***Block Diagram***



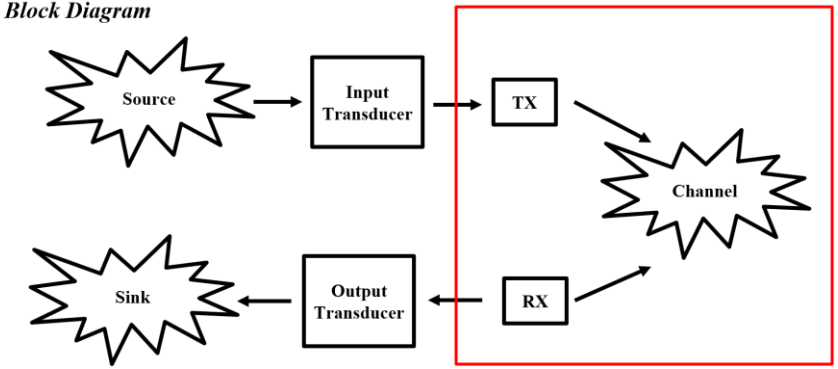
“Why Digital?

❑ ***Why Digital?***

In Analog Communication System,
We send a signal and the receiver needs to recover that particular unknown signal.

Distortion, with the same frequency band mixes with a signal is treated a part of it.

❑ *Block Diagram*

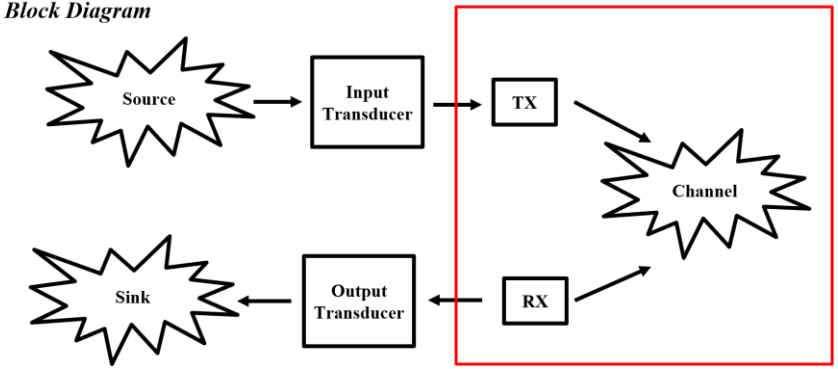


For example, telephone voice is up to 4kHz, any distortion within that band mixes with the original signal.

- ***Why Digital?***

In Digital Communication System,
We need to recognize known signals, not to recover.

- *Block Diagram*



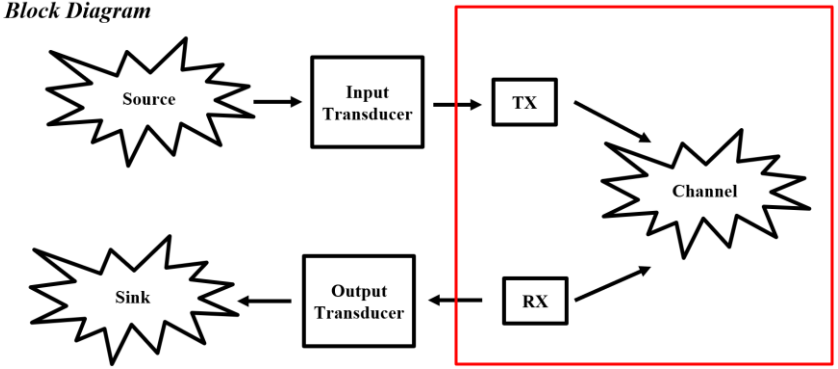


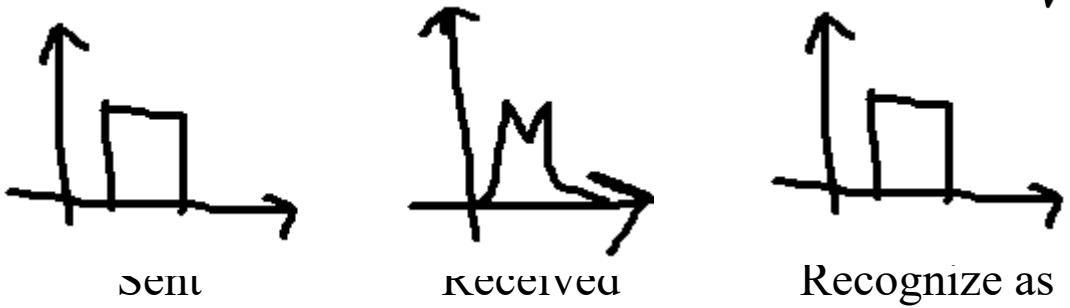
❑ ***Why Digital?***

In Digital Communication System,

We need to recognize known signals, not to recover.

❑ *Block Diagram*





Sent Received Recognize as



Gratitude

Thank you very much

Jaja Kallah Khairan